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NLP-Based Automated Essay Scoring Using BERT and Random Forest Algorithm

Marbella
Mathematics Department
School of Computer Science
Bina Nusantara University
Jakarta 11530, Indonesia
marbella@binus.ac.id

Moethia Shakira
Mathematics Department
School of Computer Science
Bina Nusantara University
Jakarta 11530, Indonesia
moethia.shakira@binus.ac.id

Alfi Yusrotis Zakiyyah
Mathematics Department
School of Computer Science
Bina Nusantara University
Jakarta 11530, Indonesia
alfi.zakiyyah@binus.edu

Kelvin A Minor
Mathematics Department
School of Computer Science
Bina Nusantara University
Jakarta 11530, Indonesia
kelvin.minor@binus.ac.id

***Abstract*—Automated Essay Scoring (AES) systems are used by teachers or lecturers to improve grading objectivity, efficiency, and scalability on large numbers of student essays. This research uses BERT for feature extraction and Random Forest for regression prediction. The dataset is from The Hewlett Foundation's AES. Data Preprocessing includes text cleaning, followed by feature extraction and model training with hyperparameter tuning. The Baseline and Normalization models give the best performance, like the value of the MSE score is 13.48 and the value of the R^2 score is 0.84. On the other hand, when comparing the models' prediction accuracy, the data normalization result showed no significant improvement and reduced model accuracy.**

Keywords—Automated Essay Scoring, NLP, BERT, Machine Learning, Random Forest

I. Introduction

Automated Essay Scoring (AES) systems are becoming increasingly important for quick and efficient grading of large numbers of student essays. Traditional grading can be challenging due to issues such as inconsistency and examiner fatigue. AES systems use Natural Language Processing (NLP) and Machine Learning (ML) to improve fairness, scalability, and predictive performance in grading [1]. One of the best NLP models for this research is BERT. BERT known for its ability to understand the context and meaning of every sentence [2].

BERT has proven to be effective in AES. For instance, Beseiso and Alzahrani [3] reported that BERT's ability to understand context led to a prediction performance with a correlation of 83.5%. Mayfield and Black [4] further improved this predictive capability to 87% with several enhancements. Similarly, Wang et al. [5] successfully employed BERT for multi-scale essay representation, achieving a prediction performance of 85.7%.

The purpose of this research is to enhance BERT's performance in AES, supported by Random Forest Regression. This study explores various factors that affect AES models, such as text characteristics like BERT embeddings, the algorithm settings like Random Forest, and preprocessing techniques like case folding and punctuation removal.

The scope of this research includes evaluating the extent to which the combination of BERT and Random Forest improves the predictive performance of Automated Essay Scoring systems compared to traditional models. It is also investigates the primary limitations of using BERT embeddings in essay scoring, particularly in relation to essay length and dataset size. Lastly, it is evaluates whether this hybrid approach can generalize effectively across different types of essays while maintaining consistency with human rater evaluations.

The AES dataset from The Hewlett Foundation is utilized for testing purposes, taking into account various variables such as essay scores, feature representation quality, essay prompt difficulty, length, and specificity. This process includes preparing text (converting to lowercase, eliminating numbers, punctuation, and excess spaces), and utilizing BERT to extract characteristics. A Random Forest model is going to be developed, fine-tuned, and assessed for reliable essay scoring and consistent results [6].

II. Literature Review

A. Automated Essay Scoring (AES)

AES systems use computational techniques to evaluate written essays, automating the grading process. To be able to predict scores, early systems like e-rater and Project Essay Grade (PEG) used statistical models and manually created features [7, 8]. Although essential, these methods struggled to grasp deep semantic meaning, prompting a shift towards more advanced NLP and machine learning approaches.

B. Natural Language Processing (NLP)

NLP techniques use computational techniques for learning and understanding human language. More complex AES systems are now accessible because of developments in NLP, such as tokenization and syntactic parsing [9]. These methods create the basis for preprocessing textual data, which allows machine learning models to comprehend and analyze essays.

C. Bidirectional Encoder Representations from Transformers (BERT)

BERT, introduced by Devlin et al., revolutionized NLP through its bidirectional training approach, capturing context from both directions of a sentence [2]. This makes BERT effective for NLP tasks, including AES, where BERT helps in accurately assessing the content and coherence of essays [10].

D. Machine Learning

Machine Learning has enhanced AES systems. Deep Learning, which is part of Machine Learning, has improved the capacity to record text patterns using Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) [11, 12]. These developments make essay grading more accurate and comprehensive.

E. Random Forest Algorithm

Random Forest is an ensemble learning strategy that improves regression performance by aggregating predictions from multiple decision trees [13]. It enhances predictive reliability and helps reduce overfitting by leveraging features extracted from embedding models [14]. Combining BERT with Random Forest presents a robust and effective approach for Automated Essay Scoring.

III. METHODOLOGY

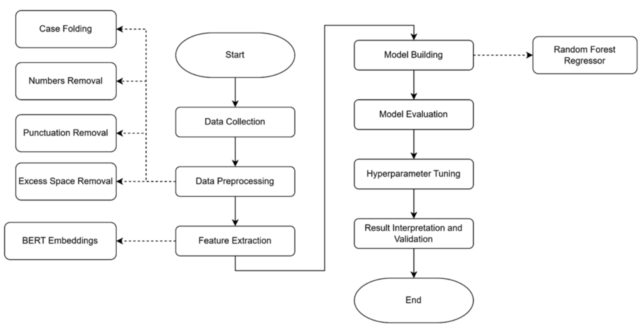

Fig. 1. Methodology Flowchart

This research focuses on improving score prediction accuracy in the Automated Essay Scoring (AES) system by utilizing two techniques: BERT for feature extraction and Random Forest for score prediction.

According to Fig.1, it is used a seven-step process. The first step is data collection and followed by data preprocessing, which includes lowercase the text, removing numbers, punctuation, and excess whitespaces. BERT embeddings are used for feature extraction to create a standardized vector for each essay.

A Random Forest Regressor is used to build the model, and hyperparameter tuning, such as Grid and Randomized Search are used to improve its performance. Mean Squared Error (MSE) and R-squared metrics are used to evaluate the model. Finally, results are interpreted and validated to ensure accurate scoring, with cross-validation used to confirm the model's reliability.

A. Data Collection

Data collection is one of the most important steps in creating the automated essay scoring system [3]. In this process, a set of essays with their scores assigned by human raters is collected. One of the most commonly used is the dataset from The Hewlett Foundation, which is also well-known as the Automated Essay Scoring dataset. This dataset has a wide coverage of various essays with their corresponding scores. This dataset is important in training and testing the scoring model, it also provides many examples to learn from and helps generalize effectively.

B. Data Preprocessing

Preparing raw text to be analyzed is an important part of data preprocessing [15]. Preprocessing includes some main steps, including converting all text to lowercase, removing numbers, punctuation, and any extra spaces. Lowercasie was used to help in developing a uniform format of texts, while removing numbers, punctuation, and unnecessary spaces, allowing us to focus on the important words

The Automated Essay Scoring dataset provided by The Hewlett Foundation contains eight distinct sets of student essays written by students from Grades 7 to 10, with each essay ranging from 150 to 550 words. All essays were manually scored by two independent human raters to ensure fairness and consistency in grading. The scoring rubrics and ranges vary by essay set but typically reflect content quality, coherence, grammar, and relevance.

Each of the eight essay sets focuses on different writing tasks and topics:

- **Essay Set 1:** Opinion-based writing on the impact of technological advancements, especially computers.
- **Essay Set 2:** Persuasive writing about censorship in school libraries.
- **Essay Set 3-6:** Source-based responses where students must interpret and analyze reading passages. Each set uses different source texts, and students are required to demonstrate comprehension by explaining details, identifying main points, or analyzing literary devices such as mood.
- **Essay Set 7-8:** Narrative and explanatory writing that is topic-focused and information-rich without using external sources.

The dataset includes training data with 28 variables (12,974 entries), validation (4,218 entries), test data (4,236 entries), and a sample submission with 4,818 entries. In this research, we only use the training data for training and testing the model.

TABLE I. DATA TRAIN DESCRIPTION

Attributes	***Description***
Essay ID	A unique identifier for each individual student'st essay
Essay Set	1-8, an ID for each set of essays
Essay	The ASCII text of a student's response
Rater1 Domain1	Rater 1's domain 1 score; all essays have this
Rater2 Domain1	Rater 2's domain 1 score; all essays have this
Rater3 Domain1	Rater 3's domain 1 score; only some essays in set 8 have this.
Domain1 Score	Resolved score between the raters; all essays have this

Attributes	Description
Rater1 Domain2	Rater 1's domain 2 score; only essays in set 2 have this
Rater2 Domain2	Rater 2's domain 2 score; only essays in set 2 have this
Domain2 Score	Resolved score between the raters; only essays in set 2 have this
Rater1 Trait1 Score - Rater3 Trait6 Score	Trait scores for sets 7-8

According to Table 1, the "Domain1 Score" serves as the dependent variable, guiding the scoring of student essays. Independent variables assess aspects like language command, clarity, and support across diverse prompts, including persuasive topics and source-dependent analysis.

C. Feature Extraction

Feature extraction converts text into numerical representations. We used BERT embeddings to capture semantic nuances. These features are then used to construct a comprehensive representation that supports accurate and consistent score prediction [3, 16].

D. Model Building

In the model building process, the Random Forest Regressor is used. The data is divided into two parts: training data and testing data, to enable the model to predict essay scores with high precision and reliability. The hyperparameter helps adjust some settings, such as '*n_estimators*', '*max_depth*', and '*min_samples_split*'. This will help to improve the model's performance and its ability to handle new data [17].

E. Model Evaluation

To evaluate the model, it is used metrics like Mean Squared Error (MSE) and R-squared are used these metrics can exactly tell how accurate and reliable the scoring is. Feature importance analysis helps in the improvement of the model and, at the same time, it gives an understanding of the model [3].

a. Hyperparameter Tuning

In hyperparameter tuning used methods, such as Grid Search or Random Search are used to fine-tune the performance of the model. These methods are used for systematically changing parameters, such as '*n_estimators*', '*max_depth*', and '*min_samples_split*'. These parameters are used to make the model more efficient and accurate in prediction [4].

b. Result Interpretation and Validation

The interpretation of the result helps us as researcher to understand how different features can influence scores. A technique called cross-validation will be used to check the model generalized to new data works well. The process validates performance on various datasets to make sure that the model is reliable and could be considered as a new example that is efficient [5].

IV. Results and Discussion

In this research, first from 28 variables at the dataset, we used 6 variables, which are 'essay_id', 'essay_set', 'essay', 'rater1_domain1', 'rater2_domain1', and 'domain1_score'. Later, we only used the 'essay' column which contains the text and 'domain1_score' as the dependent variable.

Modern language models such as BERT are trained using data in raw form without preprocessing (only cleaning is done) because they use a context-based word representation technique that relies on a more natural and direct representation of words in sentences. These models learn to understand the relationships and context of words in text without requiring additional preprocessing.

The data was split into three groups: testing (15%), validation (15%), and training (70%). The text data was embedded using BERT (Bidirectional Encoder Representations from Transformers). Distinct dataloaders were established for each set, and embeddings were produced utilizing the BERT model.

A Random Forest Regressor was initially trained on the BERT embeddings from the training set. The model was evaluated using Mean Squared Error (MSE) and R^2 Score on the validation set.

Grid Search and Randomized Search were used to tune the hyperparameters of the Random Forest model. Additionally, the dataset was normalized using StandardScaler, and the Random Forest model was retrained on the normalized data.

There are 6 models from hyperparameter tuning that focus on *'max_depth', 'max_features', 'min_samples_leaf', 'min_samples_split',* and *'n_estimators'* parameters.

TABLE II. Tuned Models' Parameters List

Model's Name	Best Parameters
Grid Search 1	{*'max_depth'*: 20, *'max_features'*: *'sqrt'*, *'min_samples_leaf'*: 2, *'min_samples_split'*: 2, *'n_estimators'*: 200}
Randomized Search 1	{*'max_depth'*: 20, *'min_samples_leaf'*: 1, *'min_samples_split'*: 5, *'n_estimators'*: 261}
Grid Search 2	{*'max_depth'*: 30, *'max_features'*: *'sqrt'*, *'min_samples_leaf'*: 2, *'min_samples_split'*: 5, *'n_estimators'*: 500}

After discussing about the parameters from TABLE II, the key insight emerged that helped to refine the search process. It was found that setting '*max_depth*': 10 consistently produced reliable performance. Additionally, increasing the number of estimators beyond 150 produced diminishing returns. These results change the development of a more focused and efficient hyperparameter tuning approach.

As a result, a new Grid Search model was introduced, with '*max_depth*' set to 10, '*n_estimators*' limited to [50, 100, 150], and smaller ranges for '*min_samples_split*' and '*min_samples_leaf*'. This targeted tuning method reduced training time, minimized overfitting, and prioritized the most effective parameters based on previous model performance.

TABLE III. Refined Models' Parameters List

Model's Name	*Best Parameters*
Grid Search 3	{*'max_depth'*: 10, *'min_samples_leaf'*: 2, *'min_samples_split'*: 2, *'n_estimators'*: 150}
Randomized Search 2	{*'n_estimators'*: 200, *'min_samples_split'*: 2, *'min_samples_leaf'*: 2, *'max_features'*: *'sqrt'*, *'max_depth'*: 20, *'bootstrap'*: False}
Randomized Search 3	{*'n_estimators'*: 500, *'min_samples_split'*: 5, *'min_samples_leaf'*: 3, *'max_features'*: None, *'max_depth'*: 20, *'bootstrap'*: True}

Here is the comparison result of Mean Squared Error (MSE) across all models, as visualized in Figure 2.

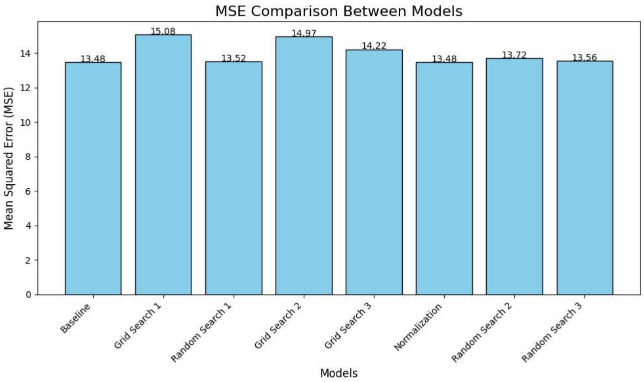

Fig. 2. MSE Comparison Between Models Bar Chart

Bar chart compares the Mean Squared Error (MSE) values for various models, helping to evaluate their performance in predicting or fitting data. The chart includes several models: Baseline, Grid Search (versions 1, 2, and 3), Random Search (versions 1, 2, and 3), and Normalization. The MSE values for each model are displayed as bars, with the height of each bar representing the error value. Lower MSE indicates better performance.

As we observed from Fig. 3, the Baseline and Normalization models tied for the lowest MSE value of 13.48, indicating they achieved the best performance among all tested models. Random Search implementations also showed promising results with relatively low MSE values, corresponding from Random Search version 1 is 13.52, Random Search version 3 is 13.56, and Random Search version 2: 13.72 Grid Search variations demonstrated higher MSE values compared to other approaches, with Grid Search version 3 is 14.22, Grid Search version 2 is 14.97, Grid Search version 1 is 15.08, representing the highest MSE among all tested models

The results suggest that hyperparameter tuning through Grid Search did not improve model performance, as evidenced by consistently higher MSE values compared to the baseline model. On the other hand, Random Search showed marginally higher MSE values than the baseline, performed better than Grid Search across all iterations, suggesting it may be a more effective hyperparameter tuning strategy for this particular case.

Lastly, the Normalization Model matched the Baseline's MSE performance, demonstrating that the normalization approach neither improved nor degraded the model's error metrics in terms of MSE.

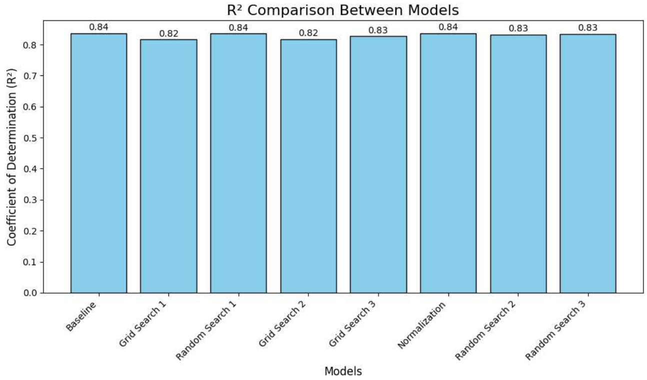

Fig. 3. R^2 Comparison Between Models Bar Chart

The bar chart shows different models by comparing their R^2 (Coefficient of Determination) values, which measure how well each model fits the data. Higher R^2 values indicate better performance, as they represent the proportion of variability in the data explained by the model.

As we observed from Fig. 4, the R^2 values across all models are consistently high, ranging from 0.82 to 0.84, indicating strong performance across all implementations. The three models tied for the highest R^2 value of 0.84 and achieved from Baseline Model, Random Search version 1, and Normalization Model. On Grid Search version 3, Random Search version 2, and Random Search version 3 all demonstrated similar performance with an R^2 value of 0.83.

On the other hand, Grid Search version 1 and Grid Search version 2 showed slightly lower performance with an R^2 of 0.82, though the difference is minimal compared to other models. Finally, the small variation in R^2 values (0.82-0.84) across all models suggests that the different approaches resulted in comparable model fits, with minimal impact on the overall predictive performance.

Furthermore, we examine the comparison between prediction scores generated by the Baseline Model and the Normalization Model, as they demonstrated optimal performance based on our analysis of Mean Squared Error (MSE = 13.48) and coefficient of determination (R^2 = 0.84) metrics.

Although traditional regression models are not typically evaluated using accuracy, this study adopts a simplified accuracy-like metric to assess model performance. A prediction is considered "correct" if the predicted score exactly matches the actual essay score. The number of exact matches is divided by the total number of test samples and then multiplied by 100 to obtain a percentage.

TABLE IV. BASELINE MODEL PREDICTION SCORE

essay_id	*domain1_score*	*predicted_baseline*
4622	2.0	3.0
6337	2.0	2.0
10387	2.0	2.0
21438	46.0	12.0

essay_id	domain1_score	predicted_baseline
21246	37.0	38.0

TABLE V. NORMALIZATION MODEL PREDICTION SCORE

essay_id	domain1_score	predicted_norm
4622	2.0	3.0
6337	2.0	4.0
10387	2.0	3.0
21438	46.0	4.0
21246	37.0	12.0

Based on Tables 2 and 3, the Baseline Model achieved 669 predictions that exactly matched the actual scores, resulting in a performance score of 34.36%. In contrast, the Normalization Model produced only 256 such predictions, corresponding to a score of 13.15%. These results indicate that the Baseline Model significantly outperformed the Normalization Model, suggesting that data normalization had an adverse effect on predictive performance in this case.

The result from this research indicated that the Baseline and Normalization models had outperformed, with a score of Mean Squared Error (MSE) is 13.48 and a score of R^2 score is 0.84. So, these models are doing a reasonably good job of predicting scores and also managing to catch the pattern in the data. But, the Grid Search method that was used to fine-tune the model's performance didn't give great performance. The Grid Search Version had a higher MSE score, which is 15.08, and lower R^2 score of 0.82.

While this method offers a straightforward interpretation, it is notably strict, since regression outputs continuous values, exact matches are rare. Therefore, this metric should be interpreted with caution.

F. CONCLUSION

This research focuses on improving the predictive accuracy of scores in the Automated Essay Scoring (AES) system by utilizing two techniques: First technique is BERT model, which technique used to extract relevant features from the essays. Second technique is Random Forest algorithm, which technique used to predict the scores.

In summary, the study concludes that Combining BERT and Random Forest is a better approach for AES than the traditional methods. This conclusion is supported by the finding data normalization does not significantly improve the model's predictive performance and may even decrease its overall effectiveness. Furthermore, the study finds that Random Search outperforms Grid Search in hyperparameter optimization for Random Forest models, highlighting the importance of efficient tuning strategis in enhancing model performance.

Based on this research, several suggestions for further development such as utilizing a more extensive dataset, To improve model generalization, it is recommended to use a more diverse dataset in terms of both the number and types of essays. This is crucial for enabling the model to handle variations in writing styles, essay lengths, and topic difficulty levels. Further, exploring other ensemble methods such as Gradient Boosting or XGBoost. Adapting the BERT model through fine-tuning on a domain-specific AES dataset can enhance feature representation and improve the accuracy of score predictions, especially for datasets within specific contexts. Also, Model performance should be analyzed more deeply across various essay categories (e.g., essay length or prompt difficulty level) to identify potential biases or model limitations.

This research can be further improved by exploring additional techniques and enhancements aimed at making the system more robust in real-world settings. To address this, the system should be implemented in real educational environments, where its practical applicability can be evaluated. Gathering feedback from educators and learners will also provide valuable insights for further refinement.

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